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Phases and microstructures of high Zn-containing Al–Zn–Mg–Cu alloys

Jun-Tao Liu*, Yong-An Zhang, Xi-Wu Li, Zhi-Hui Li, Bai-Qing Xiong, Ji-Shan Zhang

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Abstract Phases and microstructures of three high Zn-containing Al–Zn–Mg–Cu alloys were investigated by means of thermodynamic calculation method, optical microscopy (OM), scanning electron microscopy (SEM), energy dispersive spectroscopy (EDS), X-ray diffraction (XRD), and differential scanning calorimetry (DSC) analysis. The results indicate that similar dendritic network morphologies are found in these three Al–Zn–Mg–Cu alloys. The as-cast 7056 aluminum alloy consists of aluminum solid solution, coarse Al/Mg(Cu, Zn, Al)$_2$ eutectic phases, and fine intermetallic compounds η(MgZn$_2$). Both of as-cast 7095 and 7136 aluminum alloys involve α(Al), eutectic Al/Mg(Cu, Zn, Al)$_2$, intermetallic η(MgZn$_2$), and θ(Al$_2$Cu). During homogenization at 450 °C, fine η(MgZn$_2$) can dissolve into matrix absolutely. After homogenization at 450 °C for 24 h, Mg(Cu, Zn, Al)$_2$ phase in 7136 alloy transforms into S(Al$_2$CuMg) while no change is found in 7056 and 7095 alloys. The thermodynamic calculation can be used to predict the phases in high Zn-containing Al–Zn–Mg–Cu alloys.

Keywords Al–Zn–Mg–Cu alloy; High Zn; Microstructure; Homogenization

J.-T. Liu*, Y.-A. Zhang, X.-W. Li, Z.-H. Li, B.-Q. Xiong
State Key Laboratory of Nonferrous Metals and Process, General Research Institute for Nonferrous Metals, Beijing 100088, China
e-mail: b20100322@xs.ustb.edu.cn

Y.-A. Zhang
e-mail: zhangyongan@grinm.com

J.-T. Liu, J.-S. Zhang
State Key Laboratory for Advanced Metals and Materials, University of Science and Technology, Beijing 10083, China

1 Introduction

Al–Zn–Mg–Cu aluminum alloys (7000 series) are extensively used in aeronautical applications, owing to their very high strength. In the past decades, Al–Zn–Mg–Cu alloys and their process development have led to very high-strength grades, with changing the major solute elements, and with trace elements such as Zr, Cr, Sc contents [1–3]. 7055-T77 alloy with Zn content between 7.5 wt% and 8.4 wt% offers a strength increase of about 10 % relative that of 7150-T6 with lower Zn content [4]. Based on 7055 alloy, higher Zn content Al–Zn–Mg–Cu alloys were developed. 7056, 7095, and 7136 alloys with Zn content higher than 8.4 wt% are expected to replace 7055-T77 alloy [5–8]. The values of Zn/Mg are almost the same, while that of Cu/Mg of 7056 alloy is lower than that of 7095 alloy. High Zn content and different Cu/Mg ratios will produce different phases and microstructure characters in Al–Zn–Mg–Cu alloys.

According to the existing literatures, four major intermetallic phases, η(MgZn$_2$) hexagonal phase, T(Al$_2$Zn$_3$Mg$_3$), S(Al$_2$CuMg) orthorhombic phase, and θ(Al$_2$Cu) phase, can occur in commercial Al–Zn–Mg–Cu alloys below solidus temperature [9, 10]. Phase constitution and microstructure will change with the composition of Al–Zn–Mg–Cu alloys. Mondal and Mukhopadhyay [11] revealed that all of the four phases were present in the as-cast 7055 aluminum alloy. Rokhlin et al. [12] investigated the Al–Zn–Mg–Cu–Zr–Sc system with 8 wt% Zn, 2 wt% Cu, 0.3 wt% Zr, 0.3 wt% Sc, and Mg limited from 0 wt% to 8wt%, the results showed that the phases of Mg(Cu, Zn, Al)$_2$, (Cu, Zn, Al)$_{49}$Mg$_{32}$, ScCu$_{6.6-4}$Al$_{5.4-8}$ appeared. He et al. [13] found that the as-cast microstructures of 7085 alloy consist of eutectic α + T and other trace phases. It was proved that Zn and Mg played crucial role in Zn-bearing phases, and when copper was up to

2.5 wt%, most of copper would dissolve in S(Al_2CuMg) and θ(Al_2Cu) [14]. Most of the researches pay attention to the Zn-containing alloys below 8 wt%, so in this paper, the relationship among compositions, phases, and microstructures of high Zn-containing Al–Zn–Mg–Cu alloys was investigated.

2 Experimental

2.1 Alloys

Compositions of the three researched alloys are given in Table 1. The first alloy, 7056 alloy, has the highest Zn content and the lowest Cu content. The second alloy, 7095 alloy, owns an intermediate Zn content, and the highest Cu content. The third alloy, 7136 alloy, has the lowest Zn content and the intermediate Mg and Cu content. The ingots with 210 mm in diameter and 340 mm in height were cast by semi-continuous casting. Specimens were taken at the mid-radius location of top ingots and homogenized at 450 °C for different time. Thermodynamic calculation pandat software and its database were used to predict the phases. The equilibrium model was used for phase calculation. The impurity elements (Si, Fe) were not considered in the calculation.

2.2 Microstructure analysis

The microstructures of the alloys were characterized using optical microscope (OM), scanning electron microscopy (SEM), X-ray diffraction (XRD), and differential scanning calorimetry (DSC). The samples for metallographic observation were prepared through mechanical polishing and observed using Carl Zeiss Axiovert 200 MAT optical microscope. A JSM-7100F SEM working at 20 kV attached with energy dispersive X-ray (EDX) was used to observe the microstructure of the alloys. Backscattered electron (BSE) mode was used and chemical compositions of various phases in the samples were measured in energy dispersive spectroscopy (EDS). The DSC test was performed on NETZSCH STA 409C/CD different scanning calorimetry with a heating rate of 10 °C·min^{-1} over the range from room temperature to 600 °C.

Table 1 Chemical compositions of alloys (wt%)

Alloys	Zn	Mg	Cu	Zr	Si	Fe	Al
7056	9.52	2.07	1.63	0.13	<0.06	<0.08	Bal.
7095	9.30	1.90	2.55	0.13	<0.06	<0.08	Bal.
7136	8.44	2.02	2.32	0.12	<0.06	<0.08	Bal.

3 Results

Figure 1 shows the curves of phases calculated by CALPHAD method changing with temperature. It can be observed from Fig. 1a that Al_3Zr, η($MgZn_2$), α(Al), and liquid phases can be found in 7056 alloy when the temperature is between 0 and 700 °C. Besides these four phases, θ(Al_2Cu) is also found in 7095 alloy (Fig. 1b); θ(Al_2Cu) and S(Al_2CuMg) are discovered in 7136 aluminum alloy(Fig. 1c) .

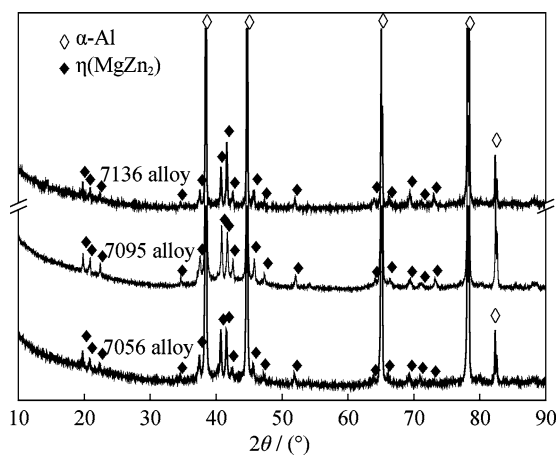

Fig. 2 XRD patterns of as-cast Al–Zn–Mg–Cu alloys

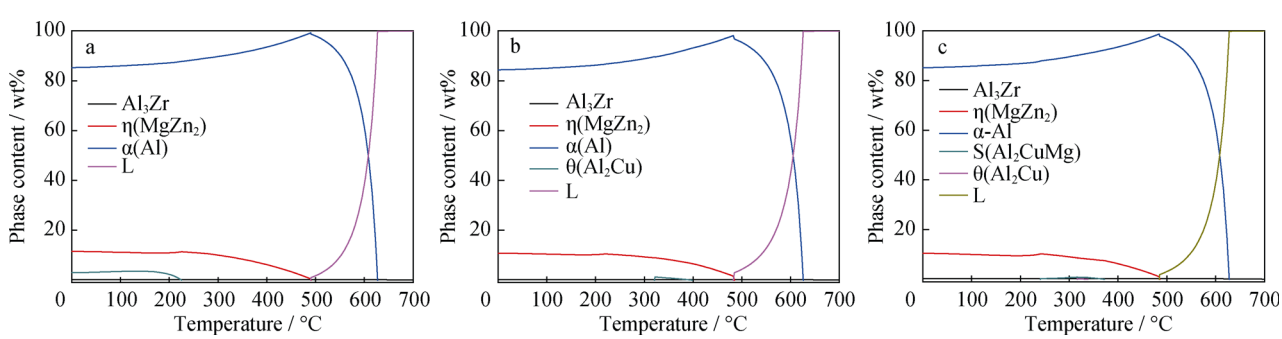

Fig. 1 Curves of phases changing with temperature in Al–Zn–Mg–Cu alloy: **a** 7056 alloy, **b** 7095 alloy, and **c** 7136 alloy

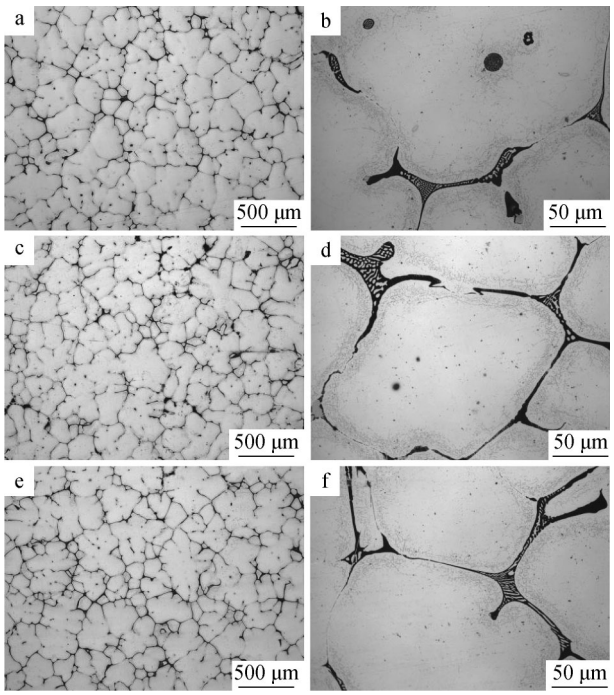

Fig. 3 OM images of as-cast alloy with different magnifications: **a**, **b** 7056; **c**, **d** 7095; **e**, **f** 7136

XRD patterns are shown in Fig. 2. It clearly can be seen that all of these three alloys consist of α(Al) and η($MgZn_2$) phases. Figure 3 shows the OM images of as-cast alloys. The three alloys exhibit similar dendritic morphology. The white areas are α(Al), and the black coarse phases form an almost network (Fig. 3a, c, and e). Lamellar structure of black coarse phases can be observed in Fig. 3b, d, and f. There are also an abundance of fine gray phases formed around the dendrites.

Figure 4 shows the typical phases of as-cast Al–Zn–Mg–Cu aluminum alloy by SEM. The dark areas are the primary solid solution, and the bright areas are the non-equilibrium eutectic phases. The EDS analysis results are listed in Table 2. The lamellar structure consists of α(Al) and AlZnMgCu quaternary phase (Points A, F, and G). And the bright coarse areas of C, E, H, and J phases are also the AlZnMgCu quaternary phase. The composition of B phase in 7056 alloy is close to stoichiometric Al_7Cu_2Fe and the compositions of D phase in 7095 alloy and I phase in 7136 alloy are close to Al_2Cu. Plenty of fine phase around the eutectic phases is taken as η($MgZn_2$) [15, 16]. Thus, according the SEM and EDS results, 7056 alloy consists of α(Al), AlZnMgCu quaternary phase, and Fe-rich phase; 7095 and 7136 alloys involve α(Al), AlZnMgCu quaternary phase, and θ(Al2Cu).

The DSC results are shown in Fig. 5. Only one endothermic peak appears around 475 °C for each curve. From the tangent rule, melting points are determined. The melting points of 7056, 7095, and 7136 alloys are 474.6, 473.3, and 475.4 °C, respectively, which are always taken as the melting of the eutectic phase (α-Al + AlZnMgCu quaternary phase). These melting temperatures correspond to the liquid phase appearance temperatures in Fig. 1. The liquid phase appearance temperature (490.0 °C for 7056 alloy, 482.5 °C for 7095 alloy, 484.2 °C for 7136 alloy) is higher than melting temperature from DSC results, respectively.

Figure 6 shows the BSE images of Al–Zn–Mg–Cu alloys homogenized at 450 °C for different time. It can be seen from Fig. 6a that fine phases around dendritic phases dissolve absolutely, and the coarse dendrite begins to dissolve when 7056 alloy homogenized at 450 °C for 6 h. The residual phase after homogenized at 450 °C for 36 h is a part of dendritic phase. Similarity can be found in 7095 alloy. However, gray phases are detected when 7136 alloy homogenized at 450 °C for 24 h. The EDS results in Fig. 6d and e are listed in Table 3. It can be found that these gray phases are close to stoichiometric S(Al_2CuMg) in composition (Points A and D). The white phases are AlZnMgCu quaternary phase. It can be concluded that S(Al_2CuMg) is transformed from coarse AlZnMgCu quaternary phase, which is similar to the results of [17].

4 Discussion

A mixture of α(Al) and lamellar eutectic phases always form in Al–Zn–Mg–Cu series alloy. Similar microstructure can be found in these three alloys. Three eutectic reactions

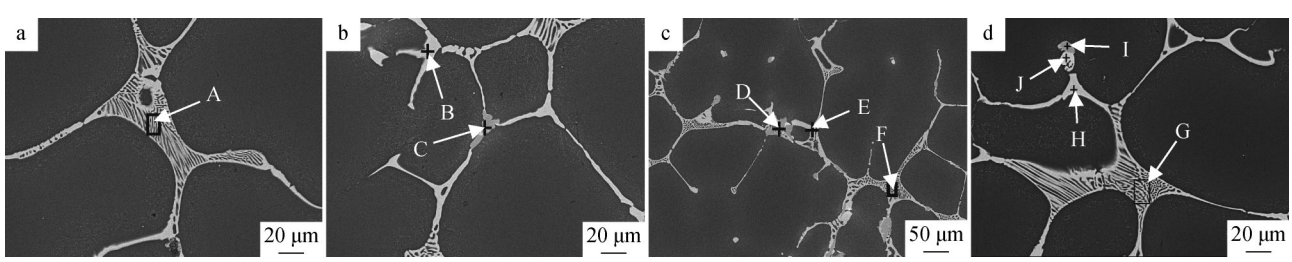

Fig. 4 Phases in as-cast 7056 alloy **a**, **b**, 7095 alloy **c**, and 7136 alloy **d**

Table 2 EDS analysis of constituent phases in Fig. 4 (at%)

Phases	Al	Zn	Mg	Cu	Fe
A	71.40	11.20	10.33	7.07	0
B	68.12	03.98	0	15.82	12.08
C	21.28	31.58	31.97	15.17	0
D	69.29	02.25	0.84	27.62	0
E	24.27	25.62	33.60	16.61	0
F	61.46	12.34	12.12	8.40	0
G	67.45	12.46	12.17	07.92	0
H	23.00	26.05	33.17	17.78	0
I	67.97	01.95	01.01	29.07	0
J	19.40	31.31	34.17	15.13	0

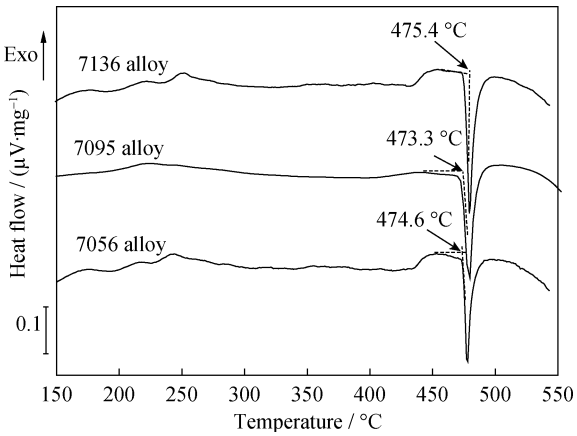

Fig. 5 DSC curves of as-cast Al–Zn–Mg–Cu alloys

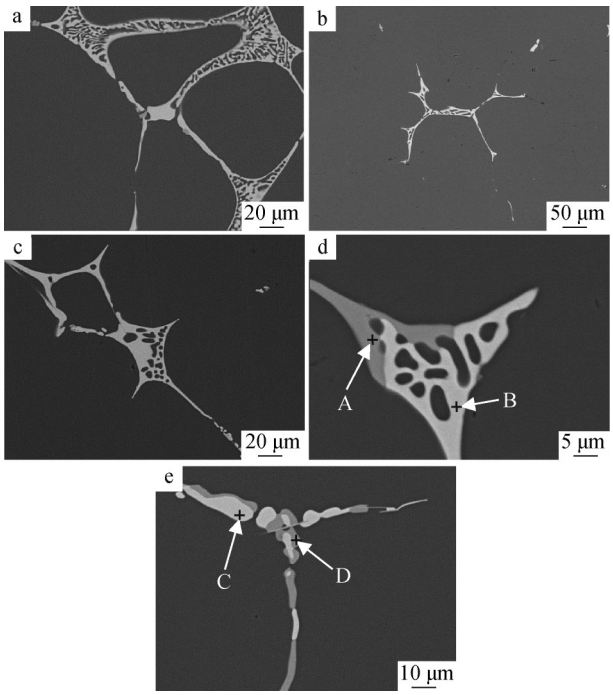

Fig. 6 BSE images showing microstructure of 7056 alloy homogenized at 450 °C for 6 h **a**, 36 h **b**, 7095 alloy homogenized at 450 °C for 24 h **c**, and 7136 alloy homogenized at 450 °C for 24 h **d**, **e**

Table 3 EDS analysis of constituent phases in Fig. 6d and e (at%)

Phases	Al	Zn	Mg	Cu
A	48.62	0.3	23.55	24.83
B	22.52	26.24	30.63	20.61
C	21.32	26.72	30.40	21.56
D	53.02	0	23.11	23.87

products of α-Al/η($MgZn_2$), α-Al/T($Al_2Mg_3Zn_3$), and α-Al/S(Al_2CuMg) are taken as the predominant eutectic structure. Besides being a part of eutectic structure, T-base and S-base phases are also present in "divorced phases" [11]. Zn/Mg ratio plays a crucial role in Zn-contained phases. It is proved that the major η($MgZn_2$) phase is formed with Zn/Mg higher than 3.0 [18, 19]. Values of Zn/Mg for alloy 7056, 7095, and 7136 alloys are 4.59, 4.89, and 4.18, respectively, and η($MgZn_2$) is the major constituent. No T-base and S-base phases can be discovered in these three as-cast alloys. Al and Cu elements will substitute for Zn at the Zn sublattice of binary η($MgZn_2$) in a wide range, which can be described as $Mg(Al,Cu,Zn)_2$. The EDS results of Points 1 and 4, in Table 2 show that the atom ratio of Mg:(Al,Cu,Zn) is about 1:2. So, it can be concluded that the AlZnMgCu quaternary phase in Fig. 4 is $Mg(Al,Cu,Zn)_2$, which is similar to the results of Refs. [11, 15–17, 20]. The area of endothermic peak shows the amount of melting phases in DSC curves. The areas of 7056, 7095, and 7136 alloys are 9.504, 9.811, and 9.324 $J\cdot g^{-1}$, respectively, increasing with the increase of Zn/Mg. This coincides with the Zn/Mg, which decides the amount of Zn-containing phases in thermodynamic calculation in Fig. 2.

The value of Cu/Mg affects Cu-rich phases. θ(Al_2Cu) appears both in thermodynamic calculation(Fig. 2) and SEM analysis (Fig. 4) of 7095 and 7136 alloys whose Cu/Mg ratios are 1.34 and 1.14, respectively. S(Al_2CuMg) is present in 7136 alloy with intermediate Cu/Mg only in thermodynamic calculation (Fig. 2). No Cu-rich phase is discovered in 7056 alloy with 0.79 Cu/Mg both in thermodynamic calculation (Fig. 2) and SEM analysis (Fig. 4). Cu will dissolve in η($MgZn_2$) phase completely when Cu/Mg is low enough. S or θ phase will first precipitates, and no change occurs during the solidification. The reasons that S(Al_2CuMg) can not be found in 7136 alloys may be the amount of S(Al_2CuMg) is too little to be discovered in

SEM or the appearance of S phase calculated only in equilibrium model. The phases found in Fig. 6d and e show that S phase can be acquired from homogenization treatment, which shows the validity of thermodynamic calculation.

5 Conclusion

In this study, three high Zn-containing Al–Zn–Mg–Cu alloys were investigated. Experiment results show that the microstructure of the as-cast 7056 aluminum alloy is composed of α(Al), α(Al)/Al/Mg(Cu, Zn, Al)$_2$ eutectic phases, and fine $MgZn_2$ phase, and microstructures of both the as-cast 7095 and 7136 alloys consist of α(Al), α(Al)/Al/Mg(Cu, Zn, Al)$_2$, fine $MgZn_2$, and $\theta(Al_2Cu)$. Mg(Cu, Zn, Al)$_2$ eutectic phases in 7136 alloy transformed into $S(Al_2CuMg)$ phase. Phases appearing in the as-cast and homogenized alloys are in accordance with the thermodynamic calculation results.

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